# Technical University of Denmark

**SPRING, 2021**
**Campus: Lyngby**

**MODEL-BASED MACHINE LEARNING**

**Mini-test 1**

**(Time allowed: 90 MINUTES)**

Name: _____
Student ID: _____

## Part 1

Consider the following factorization of the joint probability distribution of a model:

$$p(\alpha)\left(\prod_{k=1}^{K} p(\beta_k)\right)\prod_{n=1}^{N} p(z_n|x_n, d_n, \boldsymbol{\beta})\, p(y_n|z_n, \mathbf{h}_n)\prod_{e=1}^{E} p(h_{n,e}|o_{n,e}, \alpha)\,, \tag{1}$$

where $\boldsymbol{\beta} = \{\beta_1, \ldots, \beta_K\}$ and $\mathbf{h}_n = \{h_{n,1}, \ldots, h_{n,E}\}$. The variables $\{y_n\}$, $\{x_n\}$, $\{d_n\}$ and $\{o_{n,e}\}$ are observed variables (always given).

Not a very insightful way of representing a probabilistic model, right?

**1.** (20%) To gain more insights about the dependencies in the model, draw the graphical model corresponding to the factorization of the joint distribution provided above.

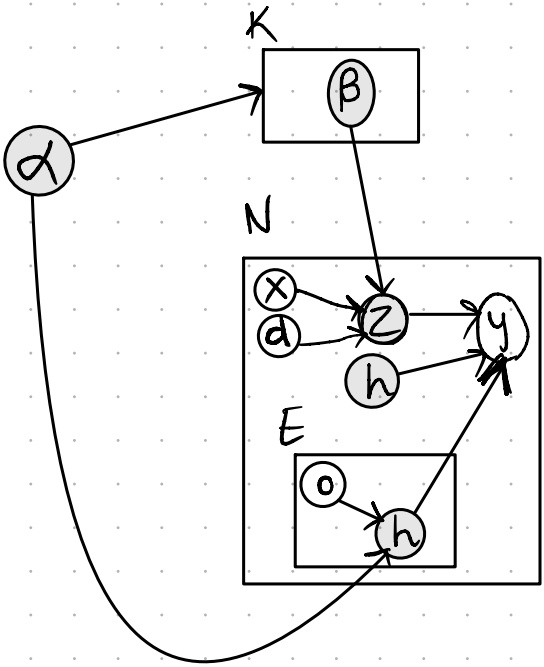
**2.** (20%) Using the D-separation algorithm, and based on the PGM that you just drew, what can you say about the following two conditional independence statements?

(a) $\alpha \perp\!\!\!\perp \beta_k|\mathbf{y}$, where $\mathbf{y} = \{y_1, \ldots, y_N\}$

(b) $\alpha \perp\!\!\!\perp \beta_k|z_2$

Hint: if the plates confuse you, consider the case where $N = 2$.
Note: if you made a mistake in drawing the PGM, that will not affect you score in this question.

1



# Part 2

Suppose that you wish to model youth pedestrian injury counts in the 603 neighborhoods of New York City for 2020.

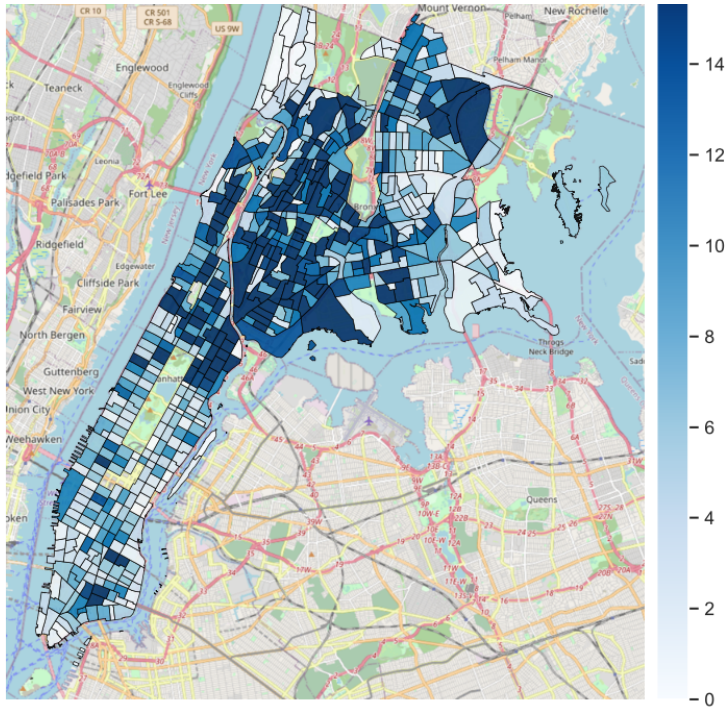



Figure 1: Youth pedestrian injury counts in New York City.

Besides the number of youth pedestrian injuries in 2020 (integer), your dataset also contains the following explanatory variables for each neighbourhood $n$:

- $\mathrm{DT}_n$ - Average **D**aily **T**raffic(float)
- $\mathrm{PS}_n$ - Proportion of households with **P**overty **S**tatus (float)
- $\mathrm{SF}_n$ - **S**ocial **F**ragmentation index (float)
- $\mathrm{PV}_n$ - Proportion of **P**rivate **V**ehicle commuters (float)

You are particularly interested in studying the relationship between the variable $\mathrm{DT}_n$ (avg. daily traffic) and the number of injuries $\mathrm{INJ}_n$ in neighbourhood $n$, so you decide to model that with a linear parametric function with some parameters $\boldsymbol{\beta}$, while for the dependency of $\mathrm{INJ}_n$ on the remaining variables you use a "black-box" modelling approach based on a neural network with parameters $\boldsymbol{\theta}$.

**1.** (20%) Write the generative process for the model described. Use appropriate distributions for all variables. If you make assumptions, state them explicitly.

**2.** (20%) Suppose now that you also have access to a binary variable $\mathrm{AA}_n$, which indicates whether a neighbourhood $n$ is predominantly African-American. You are interested in investigating whether there is racial bias in that same relationship between $\mathrm{INJ}_n$ and $\mathrm{DT}_n$. To do that, you decide to model the linear relationship between those two variables in a hierarchical way (i.e., a hierarchical model with two groups "AA" and "non-AA") and then compare the posterior distributions for $\beta_{\mathrm{AA}}$ and $\beta_{\text{non-AA}}$. Re-write the generative process of the model according to these new changes.

**3.** (20%) Draw the graphical model corresponding to the generative process that you just described in the previous question.

Note: if you made a mistake in the generative process, that will not affect your score in this question, as long as the two are coherent with each other.